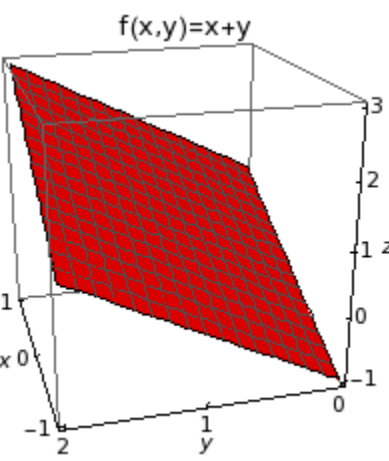
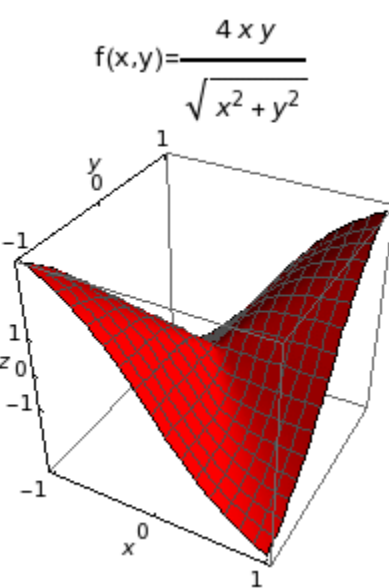


Exercício 2.1

a)

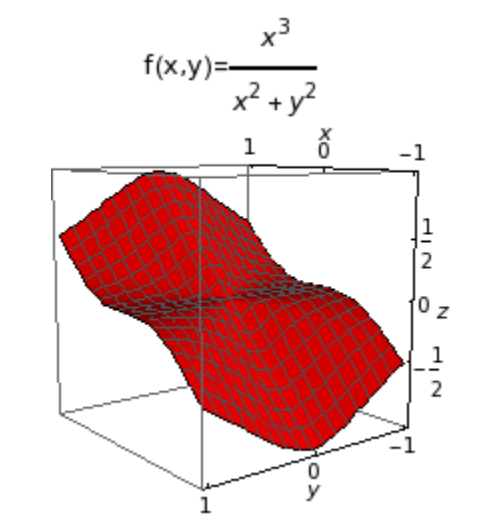


b)

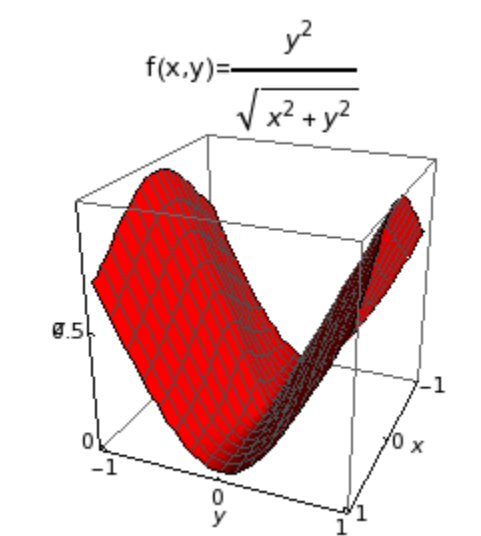


Exercício 2.2

a)

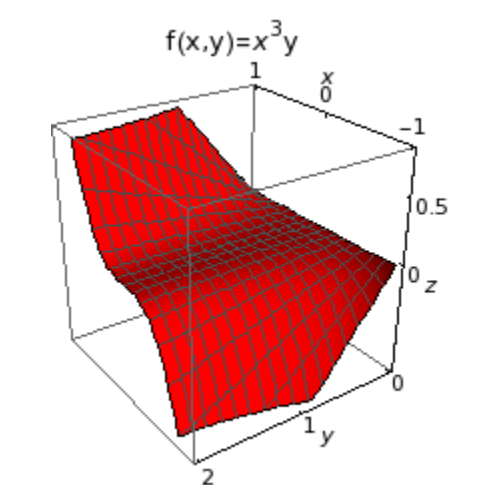


b)



Exercício 2.3

a)

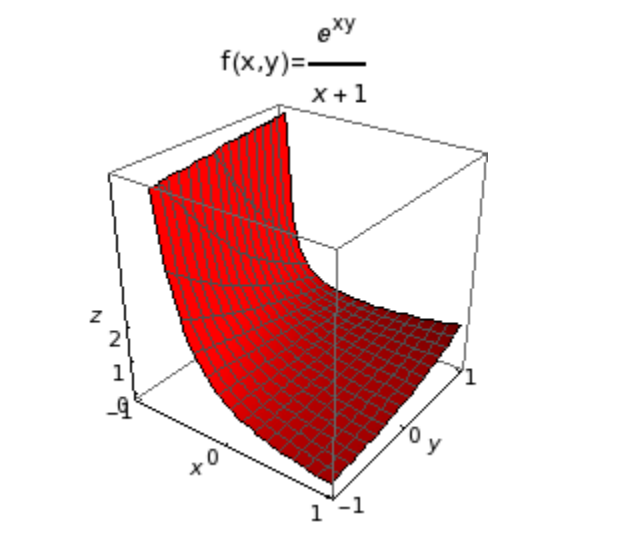


$$\lim_{(x,y) \rightarrow (0,1)} f(x,y) = 0$$

b)



$$\lim_{(x,y) \rightarrow (0,1)} f(x,y) = 0$$



$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$$

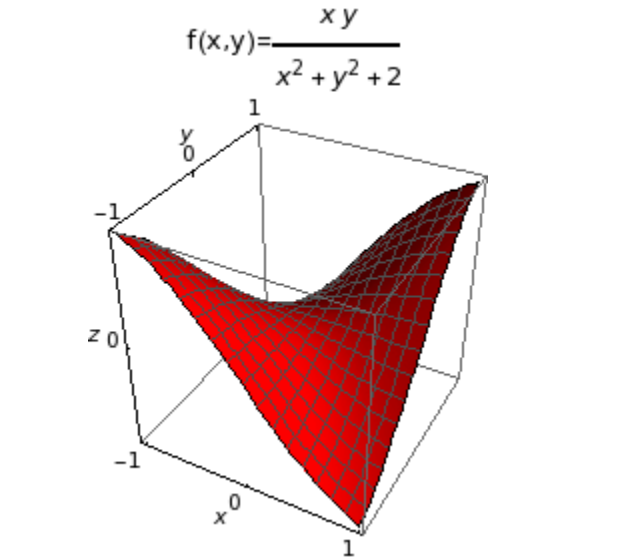
c)

$$\lim_{x \rightarrow 2} (x^2, e^x) = (1, e)$$

d)

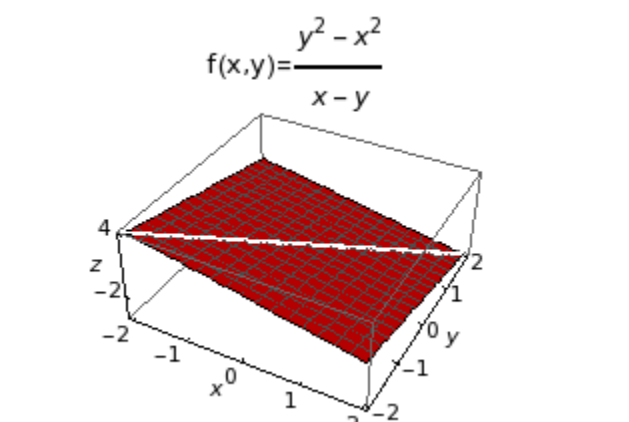
$$\lim_{(x,y) \rightarrow (0,0)} \left(\frac{\cos[x]}{x^2 + y^2 + 1}, e^{x^2} \right) = (1, 1)$$

e)



$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$$

f)



$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$$

c)

$$\lim_{x \rightarrow 2} (x^2, e^x) = (1, e)$$

d)

$$\lim_{(x,y) \rightarrow (0,0)} \left(\frac{\cos[x]}{x^2 + y^2 + 1}, e^{x^2} \right) = (1, 1)$$

e)

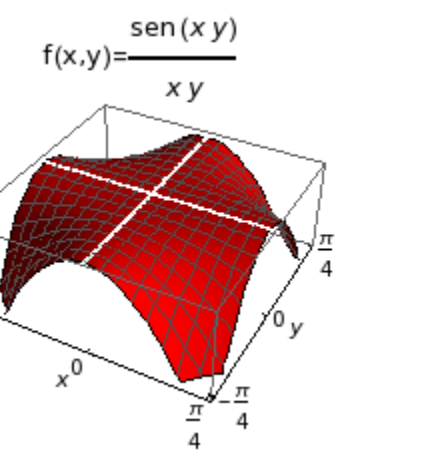


$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$$

f)

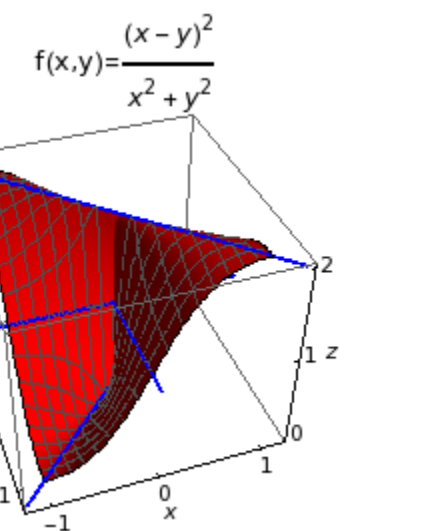


g) $f(x,y) = \frac{\sin(xy)}{xy}$
 $\lim_{(x,y) \rightarrow (1,1)} f(x,y) = -2$



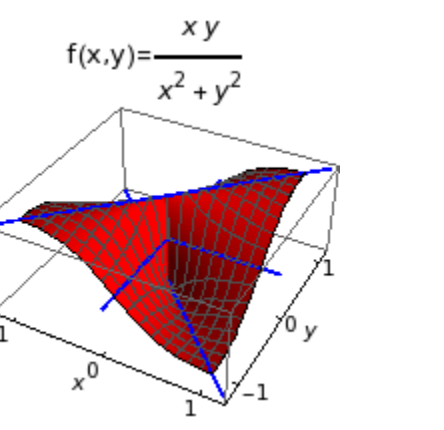
$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$

h)



Não existe $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

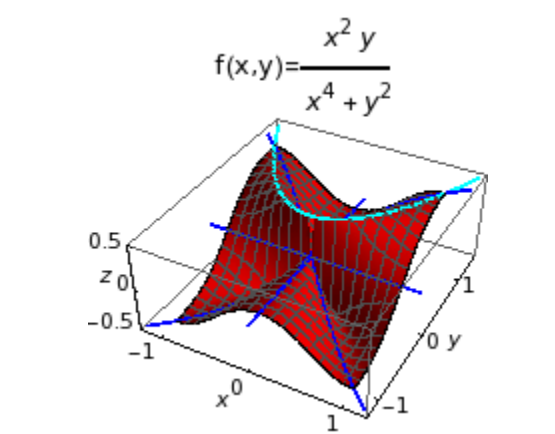
i)



Não existe $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

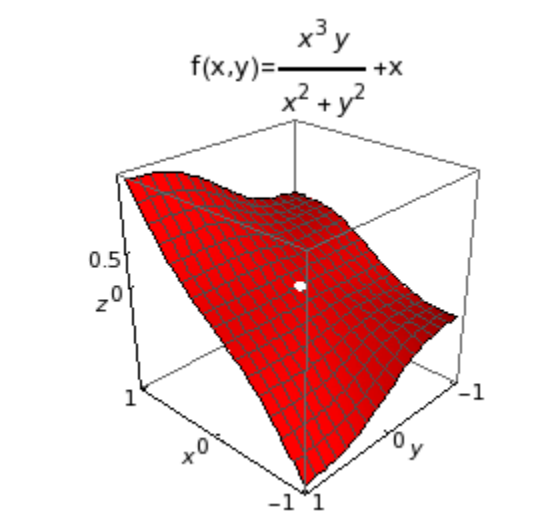
j)





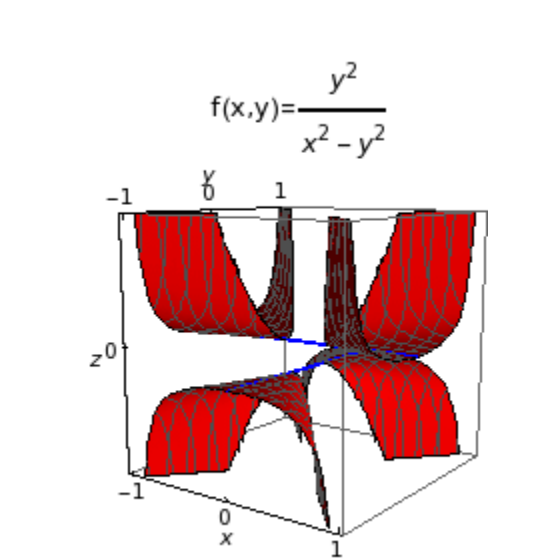
$\text{Na\o existe } \lim_{(x,y) \rightarrow (0,0)} f(x,y)$

b)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

c)



$\text{Na\o existe } \lim_{(x,y) \rightarrow (0,0)} f(x,y)$

d)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

e)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

f)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

g)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

h)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

i)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

j)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

k)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

l)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

m)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

n)



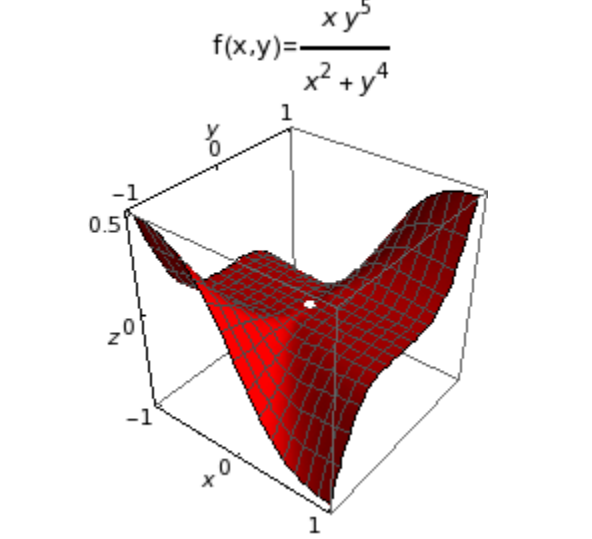
$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

o)



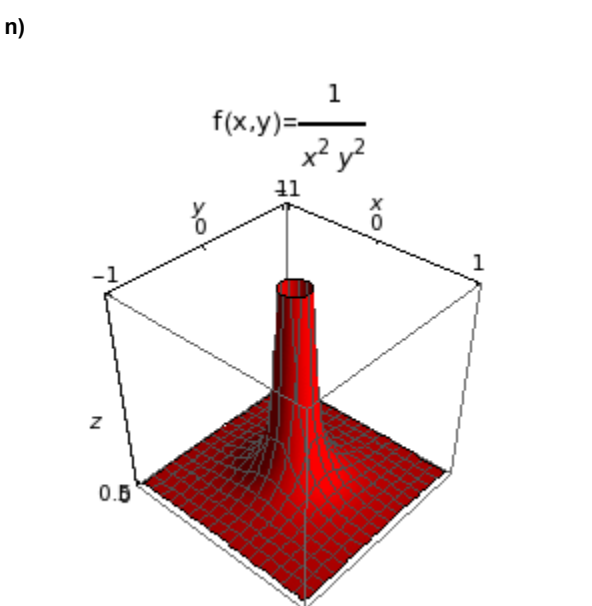
$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

p)



$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$$

n)



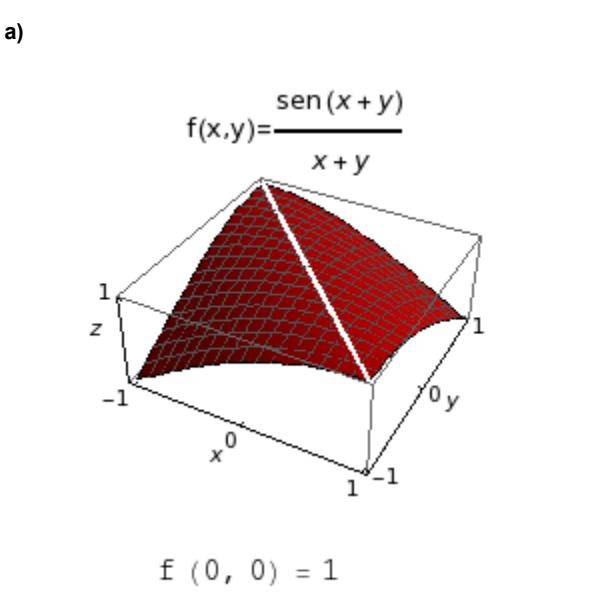
$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = +\infty$$

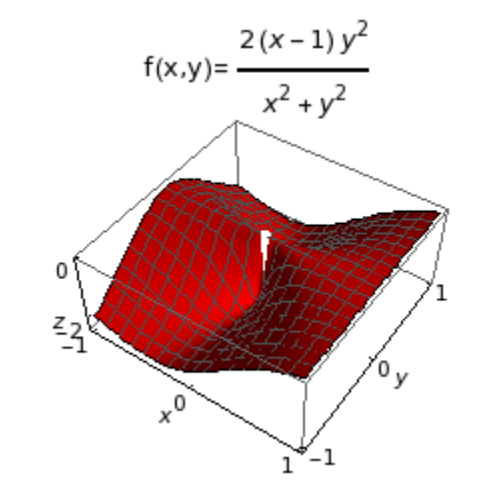
o)

$$\lim_{(x,y,z) \rightarrow (-1,1,0)} f(x,y,z) = (-1,0)$$

Exercício 2.4

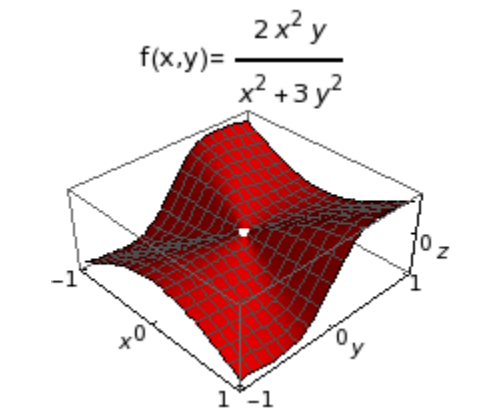
a)





Não admite prolongamento contínuo à origem

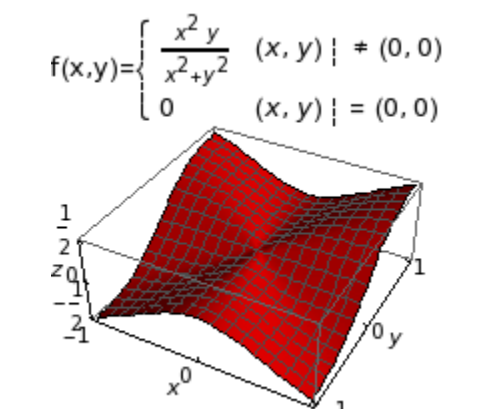
c)



$f(0,0) = 0$

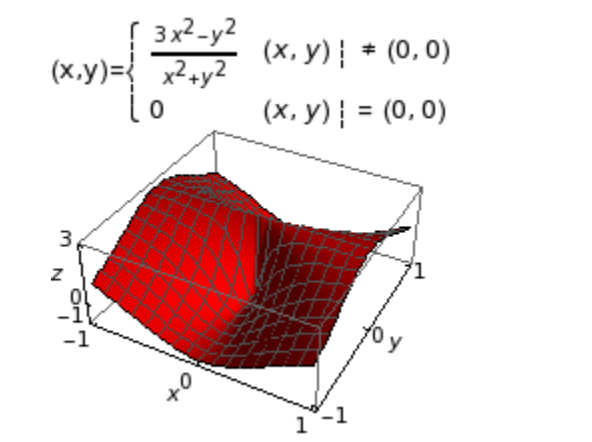
Exercício 2.5

a)



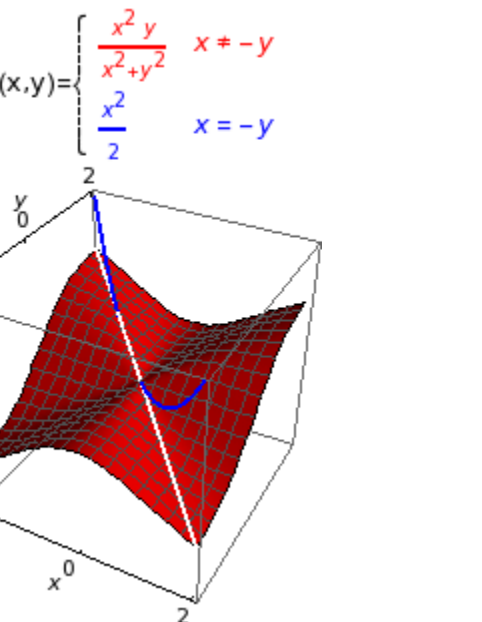
A função é contínua em $\mathbb{R}^2$

b)



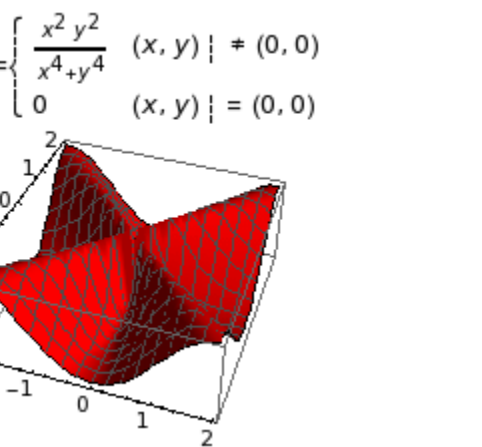
A função é contínua em $\mathbb{R}^2 \setminus \{(0, 0)\}$

e)



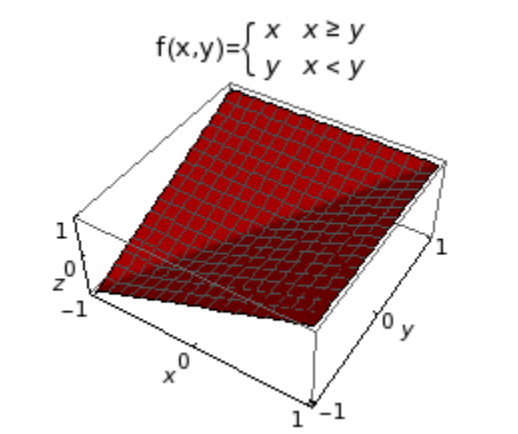
A função é contínua em $\{(x, y) \in \mathbb{R}^2 : x \neq -y\} \cup \{(0, 0), (-1, 1)\}$

d)



A função é contínua em $\mathbb{R}^2 \setminus \{(0, 0)\}$

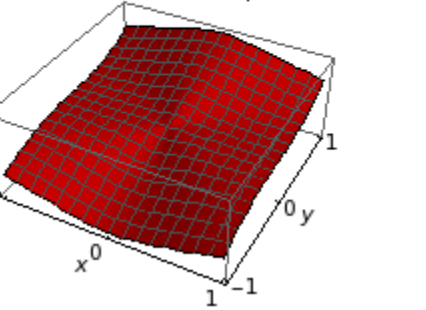
e)



A função é contínua em $\mathbb{R}^2$

h)

$$f(x,y)=\begin{cases} \frac{y^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$



A função é contínua em $\mathbb{R}^2$

g)

A função é contínua em $\mathbb{R}^3$